Title: Scene Parsing along with Image segmentation

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Dataset Description:

- Name of the dataset: ADE20K MIT License
- Annotations:

  Training: 20,210

  Validation: 2000

- Images:

  Training: 20,210

  Validation: 2000

- Test: 3352

Reference Paper links:

1. scene-parse-camera-ready.pdf
2. 1608.05442

Models used:

1) Swin transformer
2) Mask2Former
3) SegFormer

Swin transformer

1.Model Architecture

1.1 Swin Transformer

This is used as the feature extractor because of :

- Hierarchical representation learning
- Shifted window self-attention mechanism
- Efficient modelling of long-range spatial dependencies

It generates multi-scale feature maps from input images.

1.2 UPerNet Decoder

The UPerNet decoder:

- Aggregates features from multiple Swin Transformer stages that we get and
- Produces dense pixel-level semantic predictions

This Combination of Swin - UPerNet balances the local details of the images \

2. Multi-Scale Inference Strategy

for improving segmentation, **multi-scale inference** is applied:

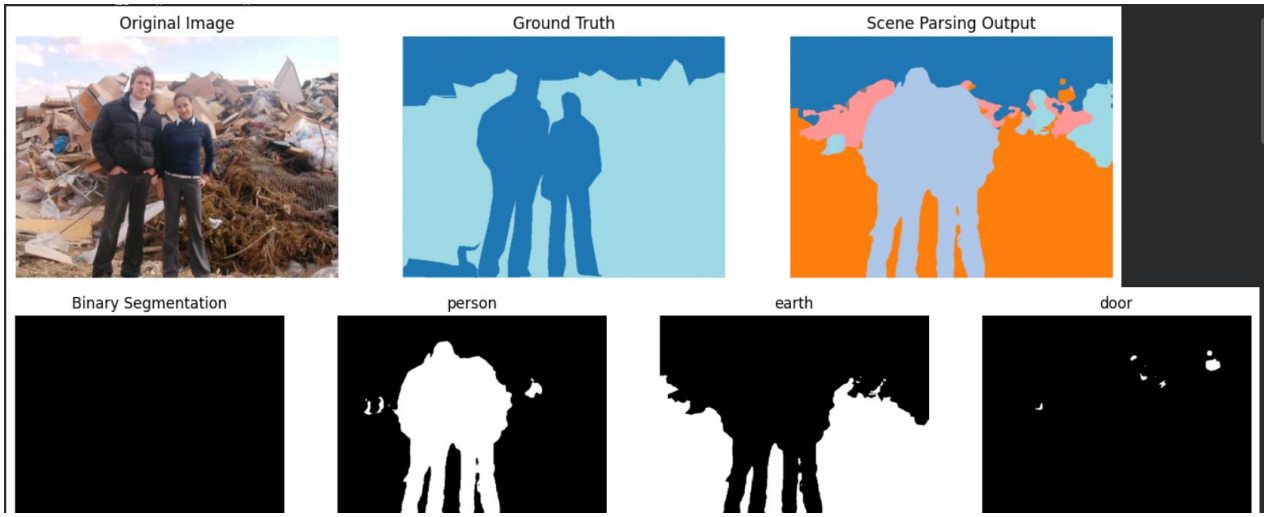
- Evry single images are resized to below scaling
  0.5×, 0.75×, 1.0×, 1.25×, and 1.5×
- Model inference will perform independently for every single scaling,
- Outputs are resized back to the original scaling (resolutions)
- Finally, we get prediction by **averaging across scales**

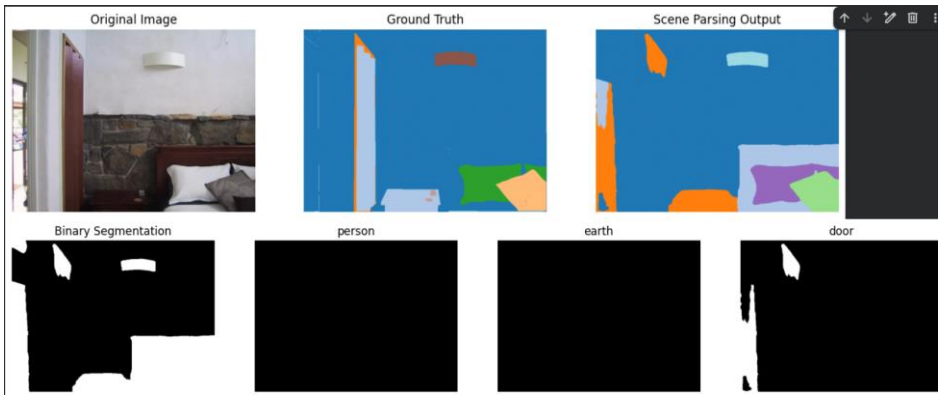
Generally, this process will (may) help in small, detailed object detection and may Increase boundary

10. Complete Workflow Summary

The overall methodology follows this sequence:

1. Load pretrained Swin + UPerNet model
2. Load and preprocess ADE20K dataset
3. Perform multi-scale inference
4. Generate scene parsing and segmentation outputs
5. Compute quantitative metrics
6. Apply 3-fold cross-validation
7. Analyze and visualize results






```
=========== BASE RESULTS ===========
Images evaluated       : 2000
Mean IoU (Primary)     : 0.5148
Parsing Pixel Accuracy : 0.8411

=========== 3-FOLD CROSS VALIDATION ===========

--- Fold 1 ---
Processing: 100%|██████████| 667/667 [06:11<00:00,  1.79img/s]
Fold 1 mIoU                : 0.5269
Fold 1 Parsing Accuracy    : 0.8438

--- Fold 2 ---
Processing: 100%|██████████| 667/667 [06:10<00:00,  1.80img/s]
Fold 2 mIoU                : 0.4962
Fold 2 Parsing Accuracy    : 0.8373

--- Fold 3 ---
Processing: 100%|██████████| 666/666 [06:11<00:00,  1.79img/s]
Fold 3 mIoU                : 0.4987
Fold 3 Parsing Accuracy    : 0.8419

=========== CROSS-VALIDATION AVERAGE ===========
Average mIoU               : 0.5073
Average Parsing Accuracy   : 0.8410
```

Mask2Former

Architecture used till now:

Input Image

↓

Scene Parsing (Mask2Former on ADE20K)

↓

Segmentation Masks (per class)

Detailed Breakdown of Process:

Load pretrained Mask2Former (Swin-L backbone)

Load ADE20K validation dataset

Run inference (predict segmentation masks)

Post-process predictions

Compare predictions with ground truth

Compute metrics (mIoU, accuracy)

Visualize results

Cross Validation of segmentation using 3fold

We are using **pre trained model, no additional training just doing inference + evaluation**

To evaluate how well a pretrained Mask2Former model performs on the ADE20K scene parsing dataset without any additional training.

"How good is Mask2Former out-of-the-box?"

You compare:

Predicted semantic masks

vs Ground-truth ADE20K labels

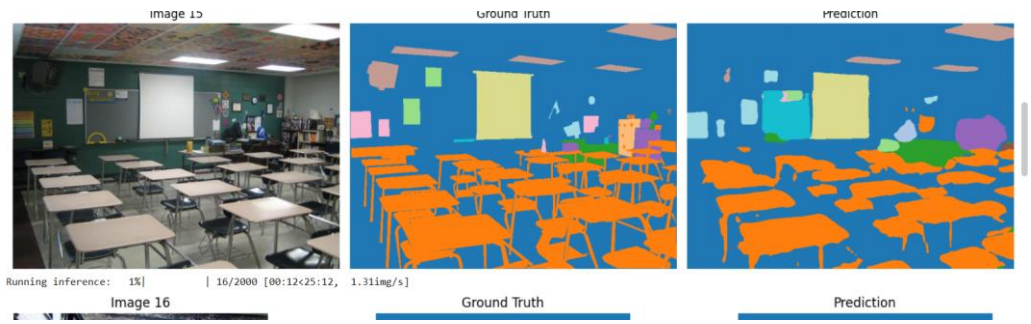
And compute Mean IoU and accuracy

This is useful for:

Model benchmarking, Baseline performance analysis, Understanding real-world generalization

Outputs:

Scene Parsing : Accuracy – 0.8393

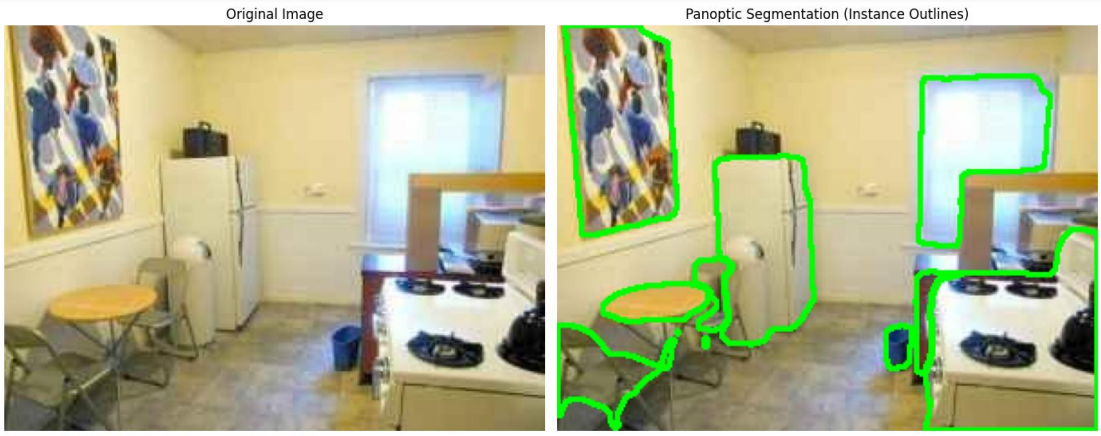



```
EVALUATION RESULTS
============================================================
Images evaluated: 2000
Mean IoU (mIoU): 0.5157
Overall Accuracy: 0.8393
Valid classes found: 150/150

Top 15 classes by IoU:
   1. Class_115           : 0.9477
   2. sky                 : 0.9466
   3. Class_57            : 0.9372
   4. bed                 : 0.8999
   5. Class_118           : 0.8836
   6. Class_81            : 0.8789
   7. Class_66            : 0.8684
   8. Class_125           : 0.8598
   9. road                : 0.8537
  10. car                 : 0.8510
  11. ceiling             : 0.8465
  12. floor               : 0.8324
  13. building            : 0.8293
  14. person              : 0.8238
  15. wall                : 0.7927

============================================================
EXPECTED PERFORMANCE (Mask2Former-Swin-L on ADE20K):
------------------------------------------------------------
From the original paper:
  Single-scale inference: 57.7 mIoU
```

```
============================================================
EXPECTED PERFORMANCE (Mask2Former-Swin-L on ADE20K):
------------------------------------------------------------
From the original paper:
  Single-scale inference: 57.7 mIoU
  Multi-scale inference: 58.1 mIoU

Note: Results on small subsets will vary.
For stable metrics, evaluate on all 2000 validation images.
============================================================

Your evaluation completed successfully!
mIoU on 2000 images: 0.5157
Result: GOOD - Close to expected performance
```

After Scene Parsing i used **Panoptic Segmentation**

Panoptic Segmentation:




```
`label_ids_to_fuse` unset. No instance will be fused.

Image 5 — Detected Objects:
  book              | Area:    507 px | Rel Volume:    507
  rug               | Area:   2977 px | Rel Volume:   2977
  television receiver | Area:   6497 px | Rel Volume:   6497
  light             | Area:   1623 px | Rel Volume:   1623
  windowpane        | Area:  11525 px | Rel Volume:  11525
  table             | Area:   3357 px | Rel Volume:   3357
  chair             | Area:   4763 px | Rel Volume:   4763
  sofa              | Area:  16274 px | Rel Volume:  16274
  table             | Area:  11690 px | Rel Volume:  11690
  plaything         | Area:   2057 px | Rel Volume:   2057
  windowpane        | Area:  12288 px | Rel Volume:  12288
  plaything         | Area:   4256 px | Rel Volume:   4256
  plaything         | Area:   2921 px | Rel Volume:   2921
```

Then did Cross Validation to segmentation

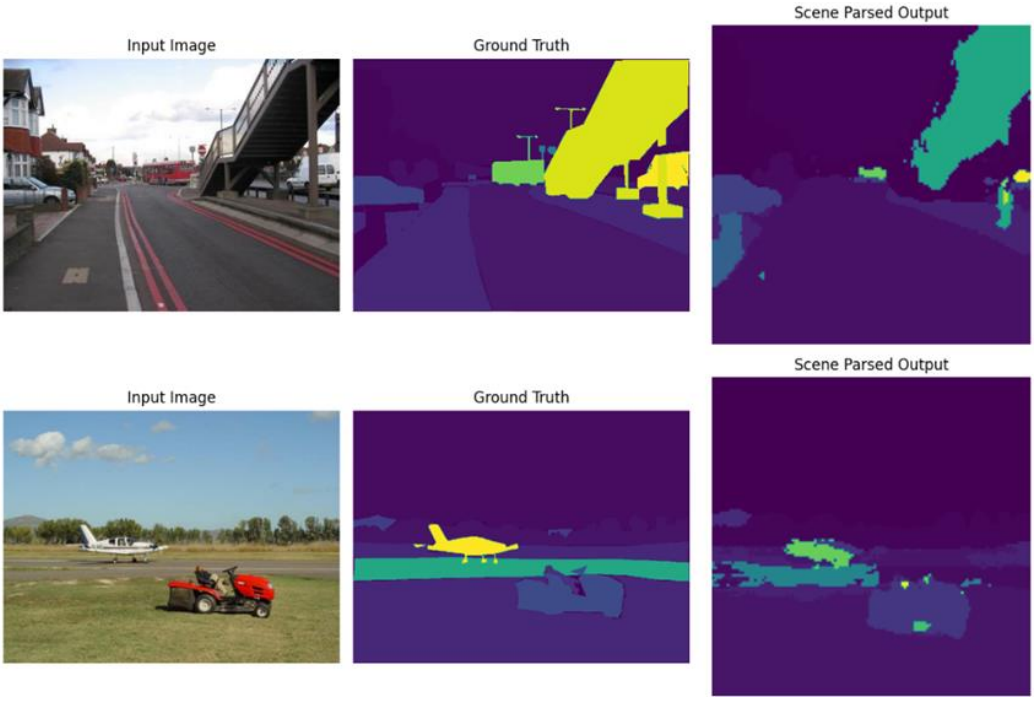
Since the model is pretrained and no learning is performed, classical k-fold cross-validation is not required. Instead, we perform a 3-fold subset-based evaluation on the ADE20K validation set to assess metric stability and robustness while maintaining computational feasibility.

```
Fold 1 mIoU       : 0.5579
Fold 1 Accuracy   : 0.8388

Fold 2 mIoU       : 0.5034
Fold 2 Accuracy   : 0.8229

Fold 3 mIoU       : 0.5317
Fold 3 Accuracy   : 0.8285

========== 3-FOLD CROSS-VALIDATION SUMMARY ==========
Mean mIoU     : 0.5310 ± 0.0223
Mean Accuracy : 0.8301 ± 0.0066
```

SegFormer for Scene Parsing:

- Used the pretrained SegFormer model and fine-tuned it on the ADE20K dataset.

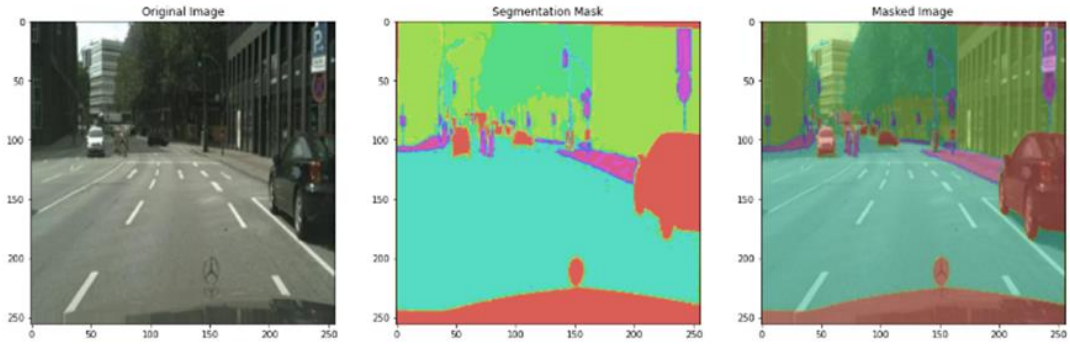
- The model was optimized using the Adam optimizer. Training was conducted for 1 epoch with a batch size of 4 per device for both training and evaluation.
- For inference, validation images from the dataset are passed through the trained model.



- Output: Pixel Accuracy = 80.86

```
{'Pixel Accuracy': np.float64(0.8085933716579413), 'Mean IoU':
np.float64(0.45847965174733996)}
```

For Image Segmentation:



- Output: Pixel accuracy = 80.68

```
tep - loss: 0.6750 - acc: 0.8068 - val_loss: 0.7489 - v
```